Classification of Matter

Answer key

A. Understanding

1. Compound vs mixture.

> A **compound** contains two or more elements **chemically bonded** in a **fixed ratio** (e.g. water, H_2O); it has new properties and can be broken down only by chemical reactions. A **mixture** contains substances that are **not chemically joined**, its composition can vary (e.g. salt water), and it can be separated by physical methods.

2. Why O_2 is an element.

> Each O_2 molecule contains **only one kind of atom** (oxygen). A compound must contain at least two different elements.

B. Classifying matter

3. Classify.

> (a) gold: **element** (b) NaCl: **compound** (c) air: **homogeneous mixture** (d) muddy water: **heterogeneous mixture** (e) sugar in tea: **homogeneous mixture** (f) CO: **compound**

C. Physical and chemical changes

4. Physical or chemical.

> (a) **physical**: change of state only (b) **chemical**: new substances (lactic acid) form; smell and texture change (c) **chemical**: wax burns to form carbon dioxide and water, releasing heat and light (the wax also melts, a physical change) (d) **physical**: still wood (e) **physical**: sugar is unchanged and can be recovered by evaporation. (Not every substance dissolves physically: calcium oxide reacts with water to form calcium hydroxide, a chemical change.)

5. Properties.

> (a) **physical** (b) **chemical** (c) **physical** (d) **chemical**

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D. Separating mixtures

6. Methods.

> (a) **filtration** (b) **evaporation** (crystallisation) (c) **distillation** (d) **chromatography** (e) **magnetism** (a magnet attracts the iron)

7. Sand, salt and water.

> 1. **Filter** the mixture: sand stays in the filter paper; salt water passes through. 2. **Wash** the sand with a little water and let it dry. 3. **Evaporate** the water from the filtrate (or heat gently until crystals form) to leave dry salt.

E. Challenge

8. Mixture vs compound FeS.

> Any two of: in the **mixture**, a magnet can pull out the iron, but not from FeS; the mixture can have **any ratio** of iron to sulfur, while FeS has a **fixed ratio** (1 Fe : 1 S atoms; about 63.5% Fe by mass); making FeS needs a **chemical reaction** (heating, which releases heat), while making the mixture does not; FeS has **different properties** from both iron and sulfur.

9. River water.

> A pure substance has a fixed boiling point at a given pressure. The river water boils 0.4 °C higher than pure water at the same pressure, so it is **not pure**: it is a **mixture** containing dissolved substances (for example salts).

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